Customer Churn Prediction using Machine Learning

Marketing Analytics (Assignment 1)

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Objective and Goal

- **Objective:**
  Predict whether a customer will churn (stop using the service) based on their behavioral, demographic, and engagement information
- **Problem Type:**
  Binary Classification (Churn = 1, No Churn = 0)
- **Goal:**

- Identify customers at risk of churning
- Help businesses improve retention strategies



Data Overview

The uses **two customer churn datasets**:

- Dataset 1: **40,000 records**
- Dataset 2: **50,000 records**
- Each dataset contains **13 features + 1 target variable (churn)**
- **No missing values** in either dataset → data is complete and ready for modeling.
- **Features include:**
- **Demographics:** age, gender, region
- **Financial behavior:** income, monthly spend
- **Customer engagement:** website visits, email click rate, support tickets
- **Marketing interaction:** campaign type, discount usage, last campaign response
- **Customer value:** tenure, loyalty score



Data Quality & Feature Separation

- **Data quality & preprocessing readiness:**
- Mixed data types:
  - Numerical: 10 features
  - Categorical: 3 features (gender, region, campaign_type)
- Clean dataset → no imputation required for missing values
- Suitable for ML pipelines with encoding + scaling



- **Key observation:**
- Dataset 2 shows **higher churn prevalence**, indicating a more imbalanced or behaviorally different customer base compared to Dataset 1.

- **Feature Separation:**
- Numerical columns
- Categorical columns

Preprocessing

- **Preprocessing Pipeline:**

1. Numerical Features

- Missing values handled using **SimpleImputer**
- Features standardized using **StandardScaler**
- Ensures all numeric variables are on a common scale for model stability

2. Categorical Features

- Missing values handled using **Most Frequent Imputation**
- Converted into numerical format using **One-Hot Encoding**
- Enables machine learning models to interpret categorical variables
- **Tool used:**

ColumnTransformer + Pipeline (Scikit-learn)



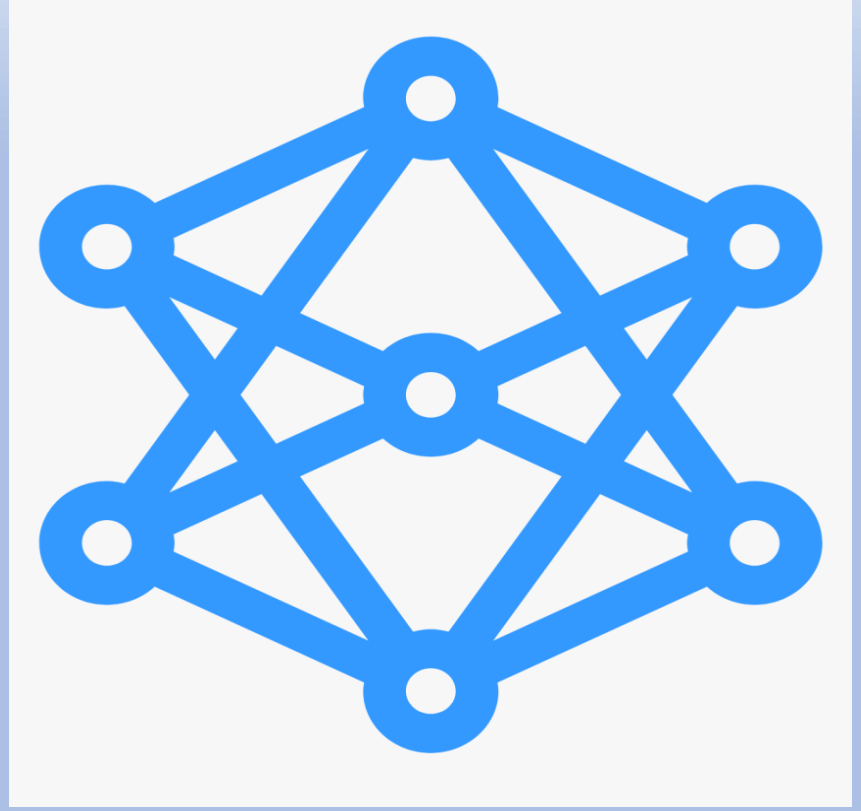
Machine Learning Models Used

- **Machine Learning Models Compared**

1. Logistic Regression
    - Linear model for classification
    - Good interpretability
2. Decision Tree
    - Rule-based model
    - Captures non-linear patterns
3. Random Forest
    - Ensemble of decision trees
    - Reduces overfitting

- **Training approach:**

- Same preprocessing pipeline used for all models
- Fair comparison ensured



Feature Importance Analysis

Methodology

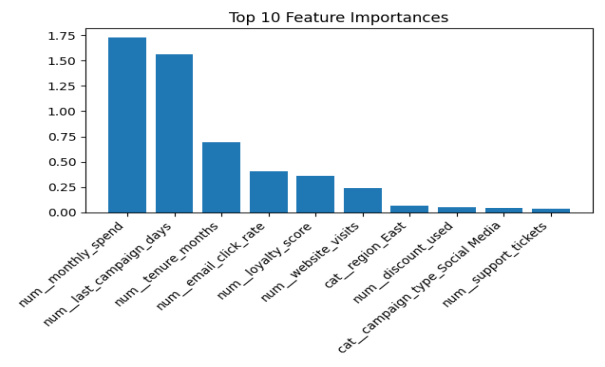
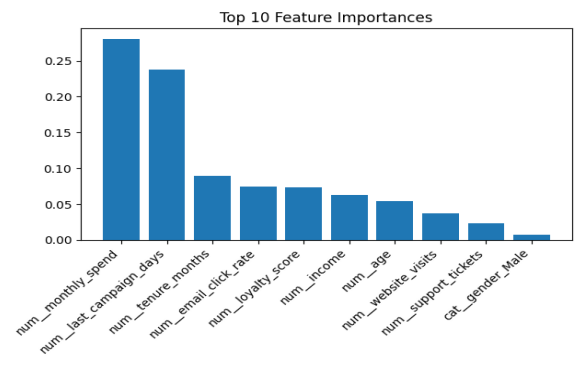
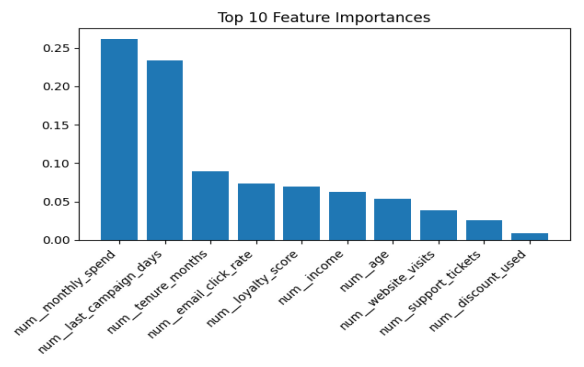
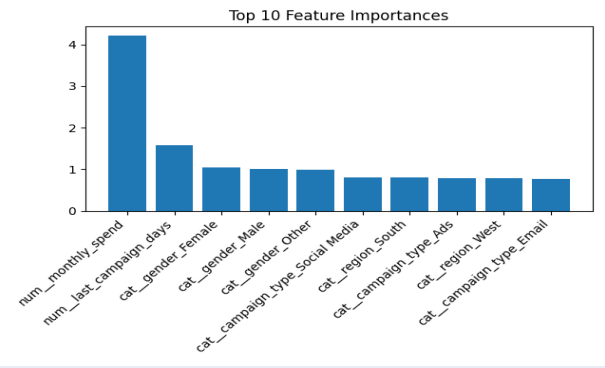
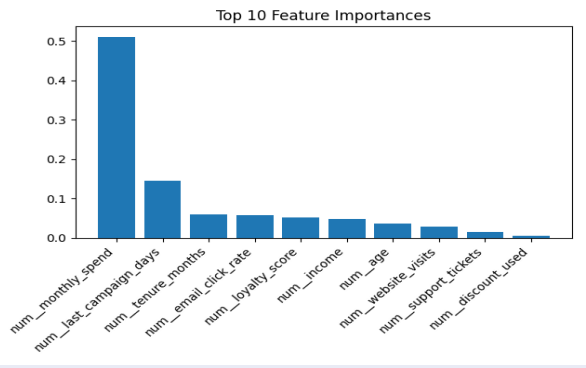
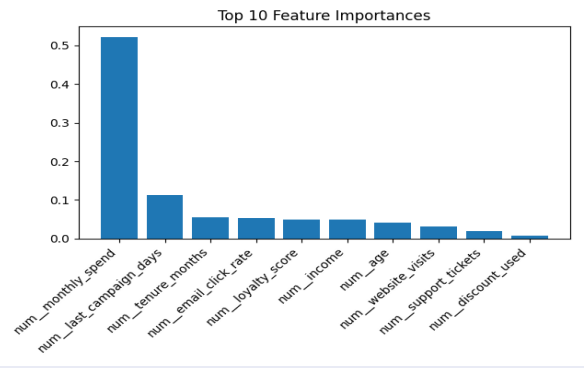
- Extracted transformed feature names from the preprocessing pipeline.

- Implemented model-specific importance extraction:
- Tree-based models -> feature_importances_
- Logistic Regression -> coef_
- Created and sorted a feature importance DataFrame.
- Visualized the Top 10 most influential features using bar plots.
- Improved model interpretability and identification of key predictive variables.

Key Insights (top 10 Features)

- **Monthly Spend** and **Last Campaign Days** are consistently the most influential features across all models and datasets.
- Logistic Regression distributes importance across multiple features, indicating broader linear relationships.
- Decision Tree and Random Forest focus more heavily on a few dominant predictors, capturing stronger nonlinear patterns.
- Dataset 2 shows more concentrated feature importance, suggesting clearer predictive structure and stronger feature separability.
- Random Forest provides the most stable and consistent feature importance rankings overall.

Top 10 Features Selected by Models

	Logistic	Decision Tree	Random Forest
DataSet 1			
DataSet 2			

Model Performance

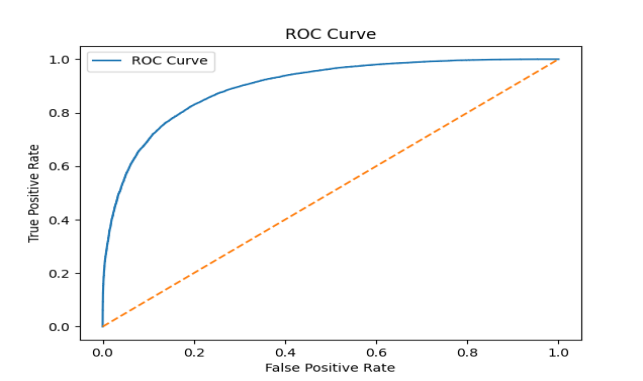
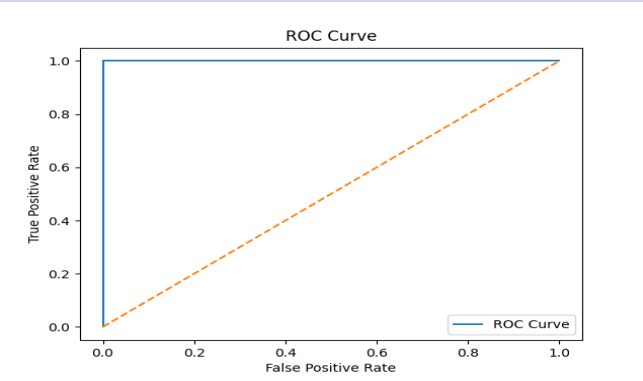
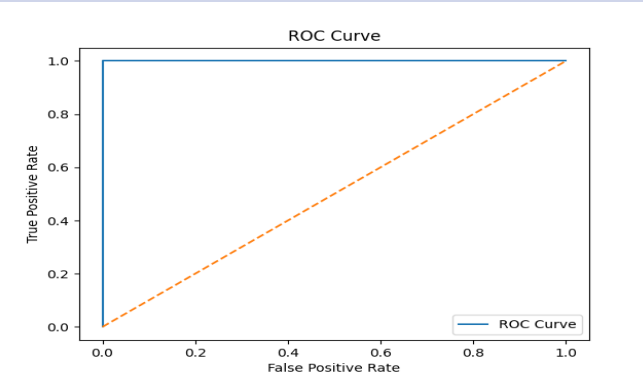
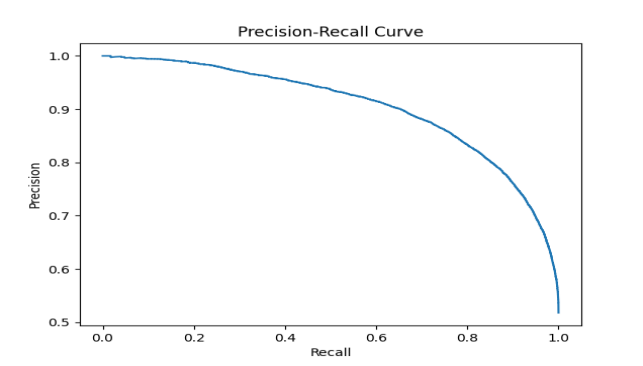
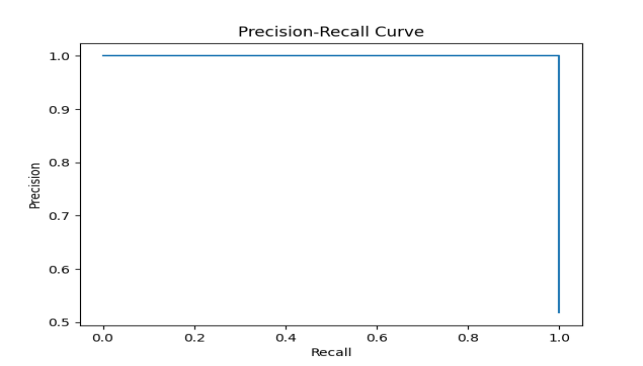
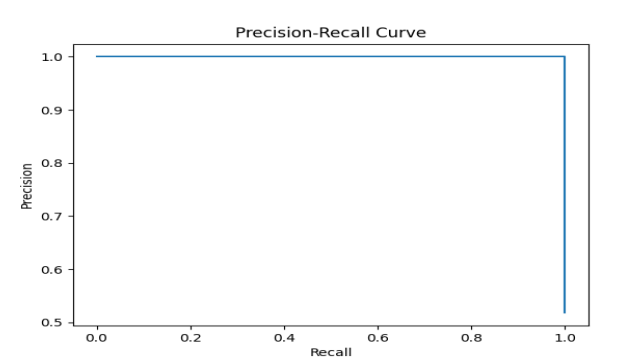
- Evaluation Metrics

- Tree-based models (Decision Tree & Random Forest) achieve **perfect scores (1.00)** on both datasets
  This is a Strong indication of **overfitting or possible data leakage**
- Logistic Regression shows **realistic and reliable performance**
- Dataset 1: Accuracy **0.815**, F1 **0.822**
- Dataset 2: Accuracy **0.942**, F1 **0.967**
- Dataset 2 is **more predictable than Dataset 1**, with consistently higher performance across all models
- Overall: **Logistic Regression is the most trustworthy model for generalization**

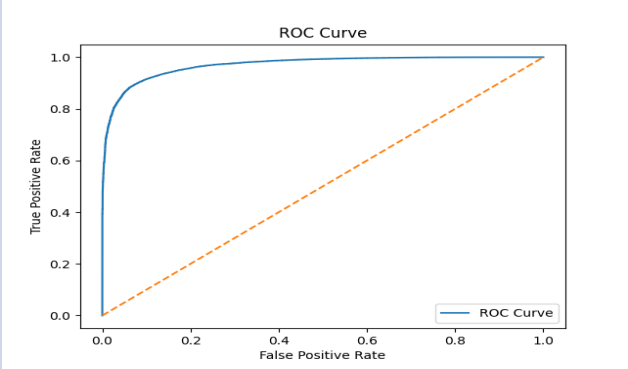
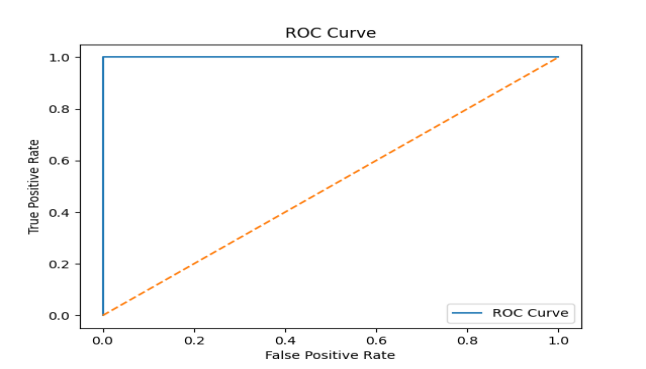
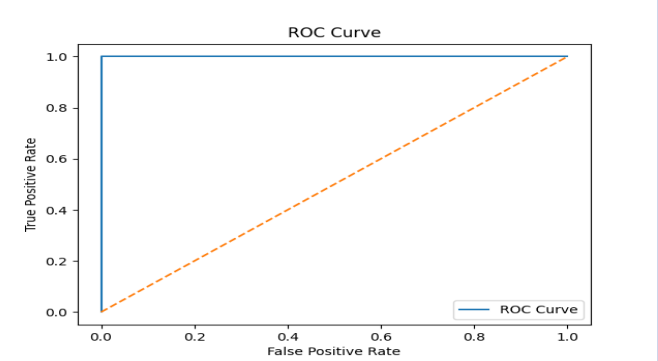
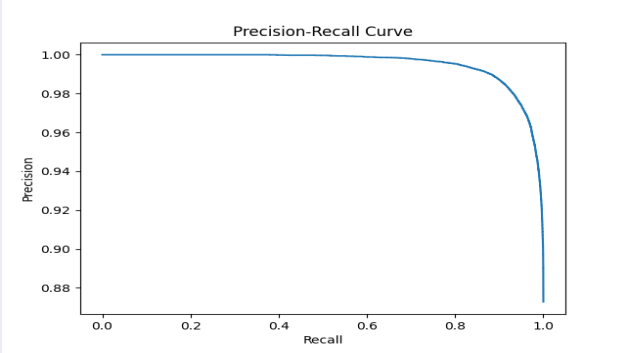
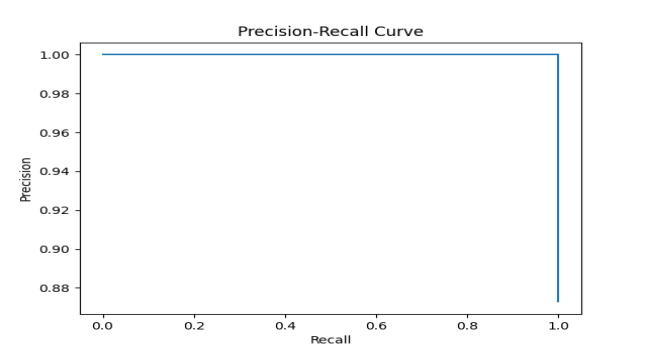
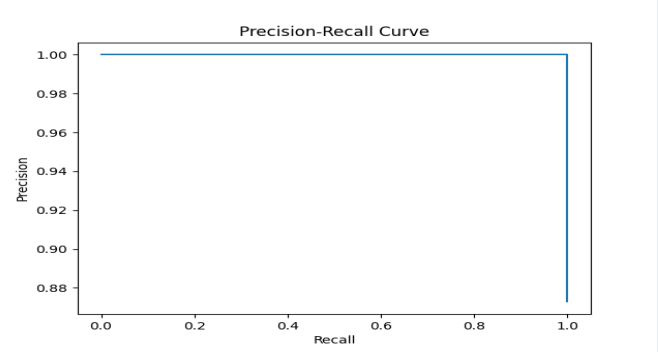
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----DATASET 1 MODEL COMPARISON---
                Model  Accuracy     F1  ROC_AUC
0  logistic_regression     0.815  0.822    0.899
1        decision_tree     1.000  1.000    1.000
2        random_forest     1.000  1.000    1.000
```

```
-----DATASET 2 MODEL COMPARISON-----
                Model  Accuracy     F1  ROC_AUC
0  logistic_regression     0.942  0.967     0.97
1        decision_tree     1.000  1.000     1.00
2        random_forest     1.000  1.000     1.00
```

ROC and Precision-Recall Curve (Dataset1)

LOGISTIC REGRESSION	DECISION TREE	RANDOM FOREST
		
		

ROC and Precision-Recall Curve (Dataset2)

LOGISTIC REGRESSION	DECISION TREE	RANDOM FOREST
		
		

Key Insight (ROC & Recall-Precision Curve)

- **Dataset 1**
- **Logistic Regression** shows strong and realistic performance with a smooth ROC and Precision-Recall curve, indicating good generalization capability.
- **Decision Tree** and **Random Forest** achieve near-perfect ROC and PR curves, suggesting extremely high classification accuracy on Dataset 1.
- The sharp "perfect" curves for tree-based models may indicate:
  highly separable classes, or potential overfitting/data leakage that should be validated carefully.
- Logistic Regression performs slightly lower but provides more stable and interpretable predictions.
- **Dataset 2**
- **Decision Tree** and **Random Forest** again achieve almost perfect ROC and Precision-Recall scores, demonstrating excellent discrimination capability.
- **Logistic Regression** also performs significantly better on Dataset 2, with ROC and PR curves much closer to the ideal top-left/top-right regions.
- Dataset 2 appears to contain clearer feature separability and more informative patterns than Dataset 1.
- **Comparative Analysis**
- Overall, Tree-based models, particularly Random Forest, achieved near-perfect classification performance on both datasets, while Logistic Regression provided strong but comparatively conservative results with better interpretability.

Conclusion

- Two customer churn datasets were analyzed to predict whether a customer will churn
- Multiple machine learning models were evaluated using performance metrics such as accuracy, precision, recall, and F1-score.
- **Logistic Regression** produced the most **trustworthy and reliable results** across both datasets, offering balanced and consistent performance.
- Although **Decision Tree** and **Random Forest** achieved **100% accuracy** on both datasets, these results may indicate **overfitting**, reducing their reliability for real-world predictions.
- The datasets were clean and complete with no missing values, which improved overall model training quality.
- Key customer behavior, engagement, financial, and marketing features contributed significantly to churn prediction.
- Logistic Regression is the recommended model for customer churn prediction in this study due to its strong generalization ability, interpretability, and dependable evaluation performance.